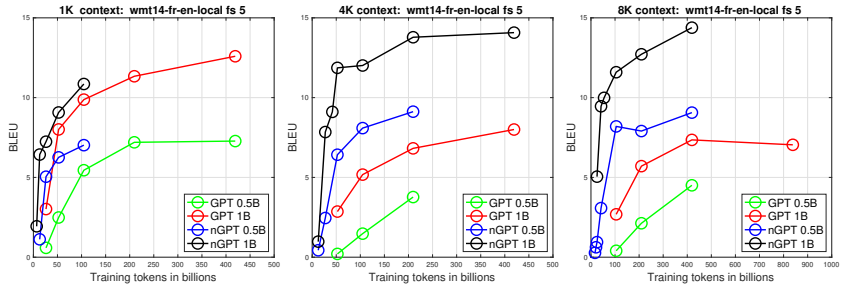

Figure 10: Performance on WMT14-FR-EN task which evaluates the ability (measured by the BLEU score of Papineni et al. (2002)) to translate a French sentence to English given five (French, English) example pairs, following the methodology of Radford et al. (2018) where it was used to benchmark GPT-2.

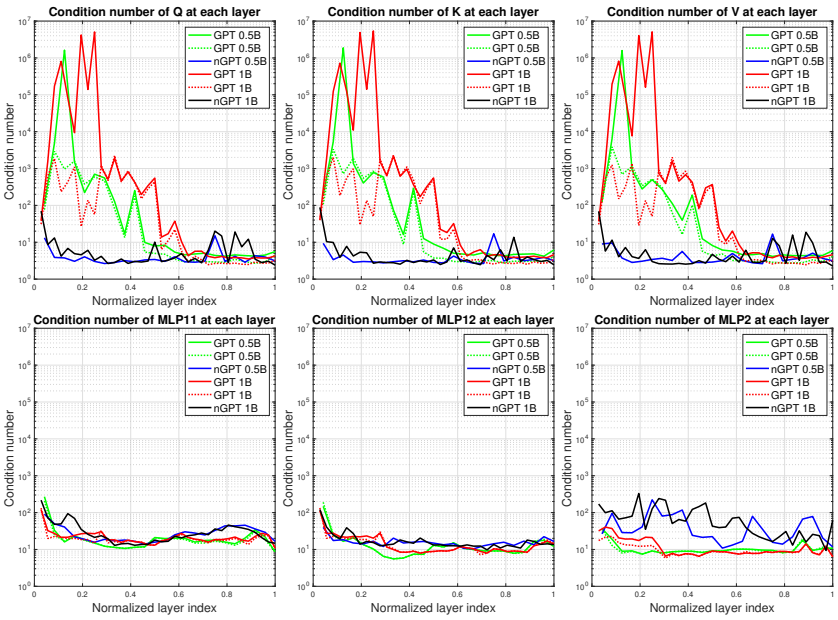

Figure 11: Median condition numbers measured for attention and MLP matrices at different layer depth (24 and 36 layers for 0.5B and 1B networks, respectively). The dotted lines are for the case when GPT's matrices are renormalized after training. Models are trained for 100k iterations.

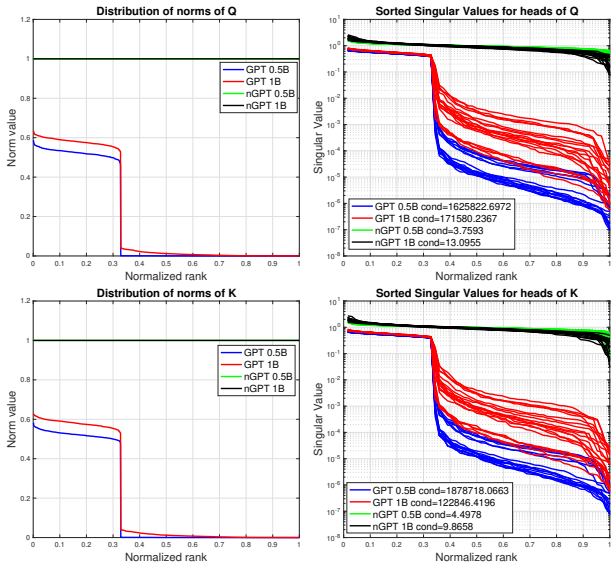

Figure 12: **Left**: Distribution of the norms of vectors forming the complete attention matrices (not per head). **Right**: Distribution of singular values for all heads. The results are shown for the 3rd layer.

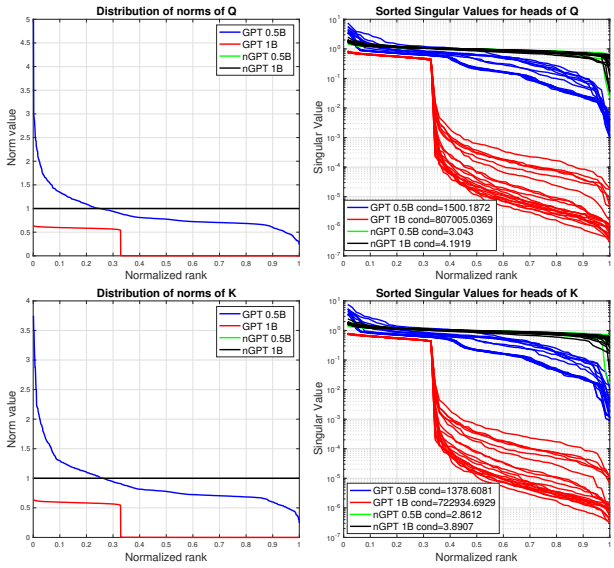

Figure 13: **Left**: Distribution of the norms of vectors forming the complete attention matrices (not per head). **Right**: Distribution of singular values for all heads. The results are shown for the 4th layer.

A.8 LENGTH EXTRAPOLATION ABILITY

We investigate the length extrapolation ability of nGPT by evaluating its perplexity on the PG19 dataset, as shown in Figure 14. In standard GPTs, perplexity tends to increase dramatically when tested on sequences longer than pre-training length. In contrast, nGPT maintains a stable perplexity range even at extrapolated lengths. This result demonstrates that nGPT can handle longer contexts without requiring any modifications to RoPE, providing a clear advantage over standard GPTs in terms of length extrapolation.

In the original version of the paper, we explored the impact of qk-normalization on performance of nGPT. Our conclusion was that the performance is equivalent but removing qk-normalization can provide computational savings. During the review process of the paper, two reviewers asked questions which made us further investigate the role of qk-normalization and context length extrapolation abilities of nGPT. This resulted in noticing that nGPT without qk-normalization shows worse extrapolation abilities (see Figure 14-Right). This difference was initially overlooked because the results of nGPT with and without qk-normalization on other tasks are practically equivalent (see, e.g., the average accuracy on downstream tasks and validation loss of both variants in Table 6). In other words, their in-distribution performance appears the same (i.e., qk-normalization does not explain nGPT's better performance) but their out-of-distribution performance is different. The difference comes from the fact that qk-normalization normalizes $\boldsymbol{q}$ and $\boldsymbol{k}$, and, thus, guarantees that their dot product is bounded in [-1,1]. Even without using qk-normalization in nGPT, the hidden state and the vectors forming $\boldsymbol{W}_q$ and $\boldsymbol{W}_k$ matrices are normalized so that their dot product is already bounded in [-1,1]. However, the results of these individual dot products (based on vectors from $\boldsymbol{W}_q$ and $\boldsymbol{W}_k$) will form $\boldsymbol{q}$ and $\boldsymbol{k}$ per head. Moreover, $\boldsymbol{q}$ and $\boldsymbol{k}$ are also affected by RoPE. The qk-normalization restores $\boldsymbol{q}$ and $\boldsymbol{k}$ vectors back to a hyper-sphere whose dimensionality is defined by the size of each head.

While the extrapolation abilities of nGPT can be partially attested to the use of qk-normalization, the gap between nGPT without qk-normalization and the baseline GPT is still substantial. Another contributing factor to the difference is the fact that many attention matrices of the baseline GPT are practically low-rank, as discussed in the main paper. This makes them less efficient in dealing with the context. Interestingly, Kobayashi et al. (2024) recently demonstrated that weight decay induces low-rank attention layers, thus confirming our observations. While the low-rank attention matrices can potentially appear due to a set of factors, we note that weight decay is not used in nGPT.

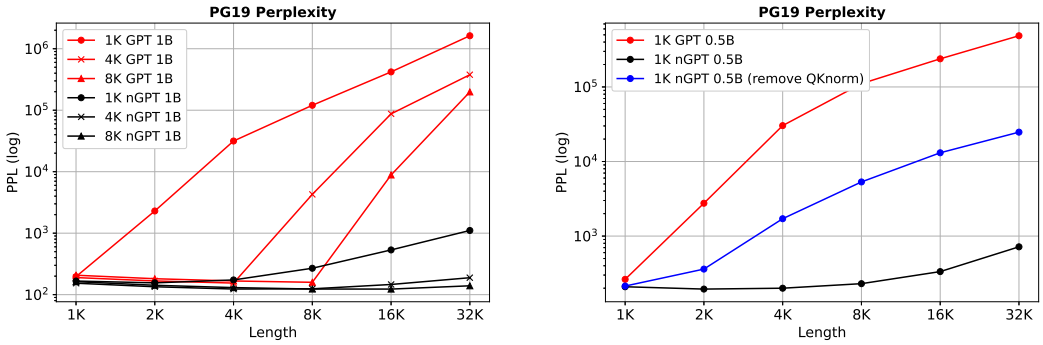

Figure 14: PG19 perplexity from 1K to 32K among different training lengths for the original GPT and nGPT (**Left**) and also with nGPT when qk-normalization is removed (**Right**).